

Clase 1

Tarea:

$$\Delta D = D \int_{T_{room}}^{T_{glnd}} \alpha(T) dT, \quad \alpha(T) = a_0 + a_1 T + a_2 T^2 \rightarrow \text{Ajuste Polinomial}$$

Integramos,

$$\int_{T_{room}}^{T_{glnd}} a_0 + a_1 T + a_2 T^2 dT = a_0 (T_{glnd} - T_{room}) + \frac{1}{2} a_1 (T_{glnd}^2 - T_{room}^2) + \frac{1}{3} a_2 (T_{glnd}^3 - T_{room}^3)$$

• Con $a_0 = 10827 \times 10^{-5}$, $a_1 = 1.11503 \times 10^{-3}$, $a_2 = -2.21004 \times 10^{-11}$

a) $a_0 (T_{glnd} - T_{room}) + \frac{1}{2} a_1 (T_{glnd}^2 - T_{room}^2) + \frac{1}{3} a_2 (T_{glnd}^3 - T_{room}^3)$

$$= 10827 \times 10^{-5} (-77 - 26) + \frac{1}{2} 1.11503 \times 10^{-3} ((-77)^2 - 26^2) + \frac{1}{3} -2.21004 \times 10^{-11} (-77^3 - 26^3)$$

$$= -1.0829 \times 10^{-3}$$

b) $a_0 (T_{glnd} - T_{room}) + \frac{1}{2} a_1 (T_{glnd}^2 - T_{room}^2) + \frac{1}{3} a_2 (T_{glnd}^3 - T_{room}^3)$

$$= 10827 \times 10^{-5} (-196 - 26) + \frac{1}{2} 1.11503 \times 10^{-3} ((-196)^2 - 26^2) + \frac{1}{3} -2.21004 \times 10^{-11} (-196^3 - 26^3)$$

$$= -2.13758 \times 10^{-3}$$